

Introduction

- Common problem in daily life – to choose the “best” route
- “best” can be different -- cheapest or fastest, always a tradeoff
- In this project, define a time budget and find the cheapest route
- Consider capacity as a limited resource. The more people using it, the higher the price would be
- Develop a piecewise linear function to represent the price floating

Model

➤ Formulation

Indices

$v = 1, 2 \dots V$	Nodes;
$e = 1, 2 \dots E$	Links;
$d = 1, 2 \dots D$	Demands;

Constants

T_d	time budget;
h_d	volume of demand d ;
$F_e(\cdot)$	price of link e ;
C_e	capacity of link e ;
t_e	time cost of link e ;
a_{ev}	= 1 if node v is originating node of link e ; 0 otherwise;
b_{ev}	= 1 if node v is terminating node of link e ; 0 otherwise;
s_d	source node of demand d ;
t_d	destination node of demand d ;

Variables

u_{ed}	binary variable corresponding to flow of demand d allocated to link e ;
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Model

➤ Formulation (contd.)

Objective

$$\text{Minimize } F = \sum_e F_e (h_d u_{ed}) u_{ed} \quad d = 1, 2 \dots D$$

Constraints

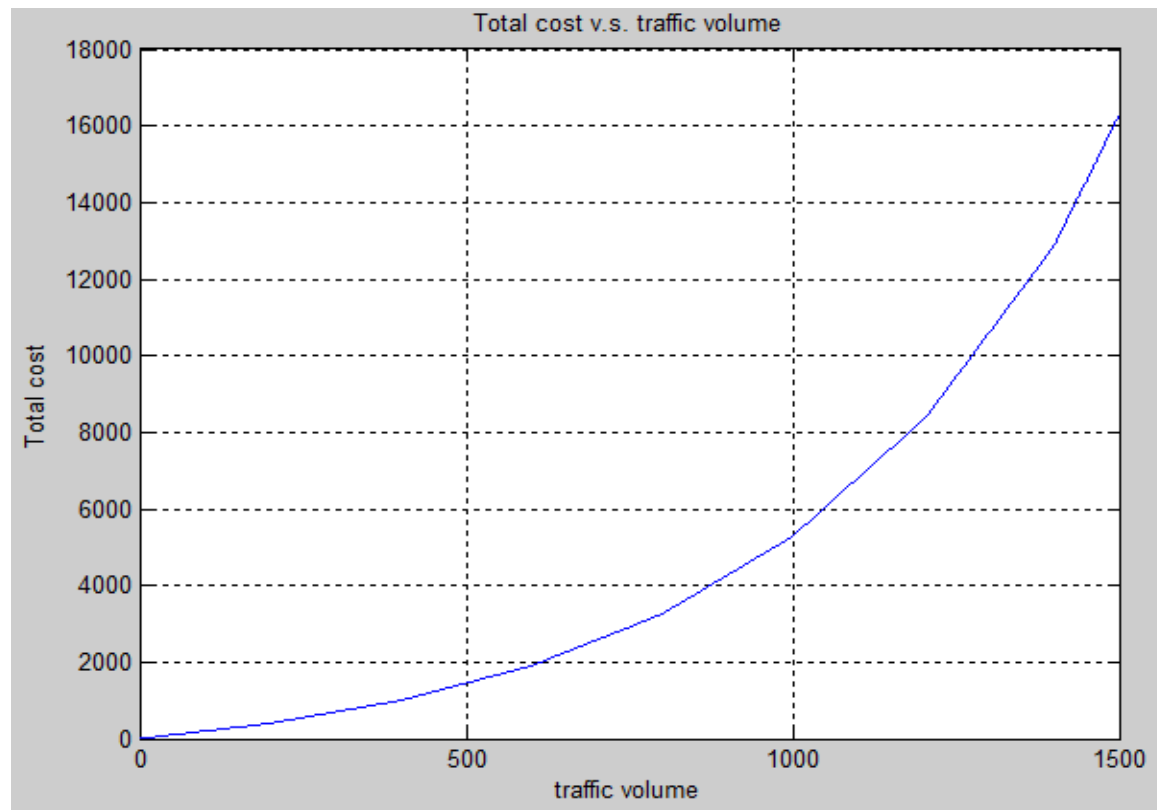
$$\sum_e a_{ev} u_{ed} - \sum_e b_{ev} u_{ed} = \begin{cases} 1, & \text{if } v = s_d \\ 0, & \text{if } v \neq s_d, t_d \\ -1, & \text{if } v = t_d \end{cases} \quad v = 1, 2 \dots V, \quad d = 1, 2 \dots D$$

$$\sum_e t_e u_{ed} \leq T_d \quad d = 1, 2 \dots D$$

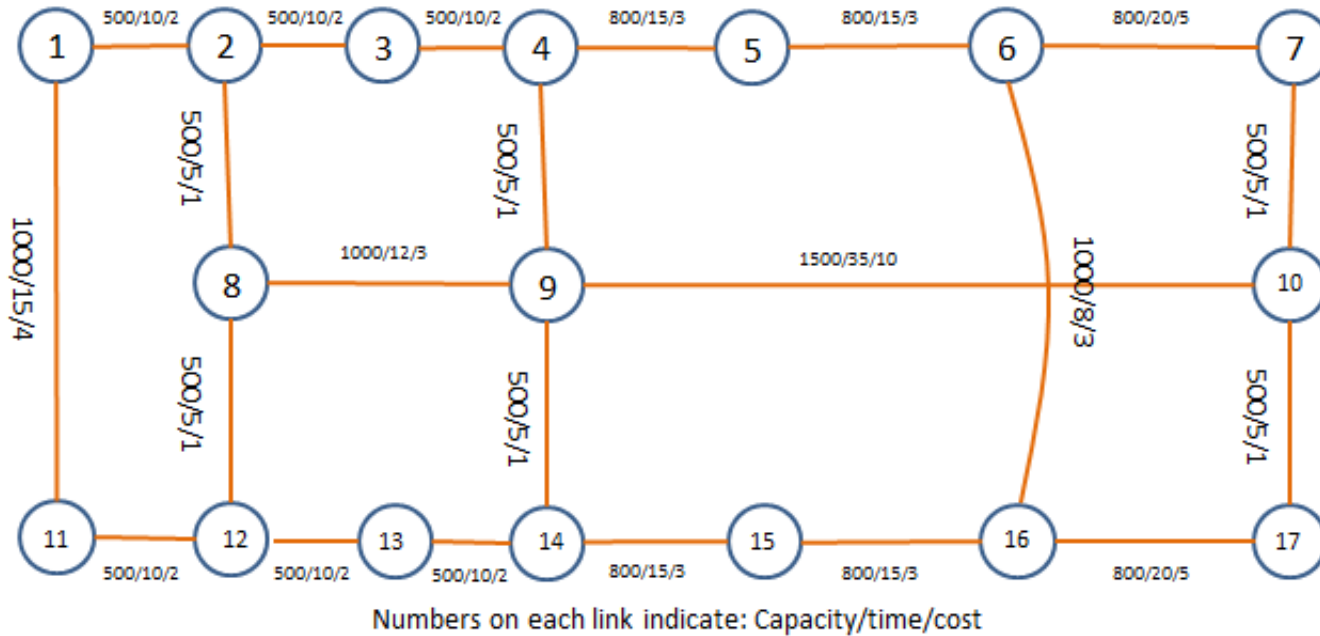
$$\sum_d h_d u_{ed} \leq C_e \quad e = 1, 2 \dots E$$

Model

➤ Cost function



Network Topology



Two directed links
between any two nodes
with same set of
parameters.

No. of nodes: 17
No. of directed links: 44

Heuristic Algorithm

Algorithm to obtain the optimal routing scheme within time budget

- (1) Explore all candidate paths from source to destination and store them in the candidate path set.
- (2) Remove the paths with travel time larger than the time budget from the candidate path set.
- (3) Remove the paths with saturated links from the candidate path set.
- (4) If there is no path available, end distributing.
- (5) Find the path with minimal cost at this iteration from candidate path set. (The unit cost of one candidate path is the sum of the unit cost of each link on this path.)
- (6) Find the maximum number of traffic can be added to the optimal path without changing the unit cost of this path. $\text{MaxFeed1} = \text{Min}\{\text{step} - \text{traffic}_{\text{link}(i)} \% \text{step}\}$. Since in this model the unit price of each link will change every 200 people, $\text{step} = 200$.
- (7) Find the maximum number of traffic can be added to the optimal path without exceeding any of its link capacity. $\text{MaxFeed2} = \text{Min}\{\text{capacity}_{\text{link}(i)} - \text{traffic}_{\text{link}(i)}\}$
- (8) Set the actual feed to be: $\text{Feed} = \text{Min}\{\text{demand volume}, \text{MaxFeed1}, \text{MaxFeed2}\}$ and add it to the optimal path.
- (9) Update the unit price of each link on the optimal path.
- (10) Demand volume = demand volume – Feed. If demand volume equals to zero, end. Otherwise, go to step (3).

Heuristic Algorithm



➤ **Example 1. Effect of Time Budget**

[illegible]

Small time budget leads to fewer feasible solutions.

C++ Results

- Example 2. Multi-cycle process for large demand volume.

[illegible]

Divide large demand into small portions to achieve overall minimal cost.

➤ Example 3. Dealing with saturated links.

Remove paths with saturated links from the feasible solution set.

Remove paths with saturated links from the feasible solution set.

➤ **Example 4. No feasible solution.**

The cause of no feasible solution may be tight time budget or no available links.

AMPL Solution

➤ From 1 to 10, demand = 500, time budget = 75

Iteration	Path (node)	Cost before updating	Volume	Total cost	Cost after updating
1	1->2->8->9->10	16	200	3200	24
2	1->2->3->4->9 ->10	23	200	4600	34.5
3	1->2->8->9->10	33	100	3300	33

Verified AMPL can also solve this floating prices problem.

Comparison

Compare two methods:

- Optimal Routing
- Naïve Routing

User's Input:

	Source	Destination	Time Budget	Demand Volume
Demand 1	Node 1	Node 10	75	500
Demand 2	Node 11	Node 4	40	500

C++ Results

[illegible]

C++ Results

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11->12->8->9->4    traffic volume: 100    total cost on this path: 1000
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Comparison

C++ Results

Path	Traffic	cost	Total Cost
11->12->13->14->9->4	200	1600	5000
11->12->8->2->3->4	200	2400	
11->12->8->9->4	100	1000	
Path	Traffic	cost	Total Cost
1->2->3->4->9->10	200	4600	11100
1->2->8->9->10	300	6500	

Total overall cost:
 $F = 16100$.

Naïve Results

Path	Traffic	cost	Total Cost
11->12->13->14->9->4	500	1600	5800
Path	Traffic	cost	Total Cost
1->2->8->9->10	500	4600	11800

Total overall cost:
 $F = 17600$.

Conclusion

- The heuristic algorithm can find the cheapest path at the best effort, and the performance is better than Naïve algorithm
- Further study: find the best step etc..